NAVEENRAJ S

Electronics and Communication Engineering

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Career Objective

Electronics and Communication Engineering student focused on VLSI Design, RTL Development, and FPGA-based systems. Hands-on experience with Verilog HDL, SystemVerilog and FPGA-based DSP implementation.Interested in digital hardware design, functional verification, and hardware acceleration.

Education

B.E. – Electronics and Communication Engineering 2024 – 2028
Bannari Amman Institute of Technology, Sathyamangalam, Tamil Nadu
CGPA: 7.97

Higher Secondary – Class XII 2023
Saraswathi Matric Higher Secondary School, Valapady, Tamil Nadu
Percentage: 89%

Technical Skills & Areas of Interest

Programming	Verilog, SystemVerilog, C, Python (Basic)
Digital Design	RTL Design, Digital Logic Design, FSM, Counters, Registers
VLSI / FPGA	FPGA Design, Functional Verification, CMOS Fundamentals
Processor Design	RISC-V, PicoRV32, Processor–Accelerator Integration
DSP	FIR Filters, Moving Average Filter, Digital Signal Processing
Areas of Interest	VLSI Design, RTL Verification, FPGA-Based Systems, DSP

Tools Used

Xilinx Vivado, QuestaSim, ModelSim, Icarus Verilog, GTKWave, Cadence Virtuoso, Synopsys Tools

Projects

Design and FPGA Implementation of an FFT-Based Industrial Signal Processing System for Machine Fault Detection Using ML [GitHub]

- Developed an RTL-based signal processing system to analyze industrial signals using Fast Fourier Transform (FFT).
- Focused on frequency-domain feature extraction for machine fault detection using machine learning techniques.

PicoRV32-Based DNA Sequence Analyzer [GitHub]

- Integrated a PicoRV32 RISC-V processor with a custom Verilog accelerator for DNA sequence matching and motif detection.
- Applied the Smith–Waterman algorithm to identify local similarities between DNA sequences.

Moving Average FIR Filter Design and FPGA Implementation Using MATLAB and Xilinx Zynq-7000 [GitHub]

- Implemented a parameterized Moving Average FIR Filter using Verilog HDL on Xilinx Zynq-7000, integrating MATLAB-generated samples for FPGA-based signal processing.
- Evaluated hardware scalability across different filter window sizes by comparing LUT and flip-flop utilization, and verified FPGA outputs against MATLAB reference results

Experience

Student Mentor – First-Year Students 2025

- Mentored first-year engineering students in fundamental analog and digital electronics concepts through technical guidance and practical explanations

Training & Certifications

- VLSI Training – Maven Silicon (Pursuing)

Achievement

- Secured 3rd Place in RISC-V Make-a-thon at Bannari Amman Institute of Technology for developing a PicoRV32-powered hardware-accelerated DNA sequence analysis and motif detection system.